# Stats 596 – Homework 2

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1. **Exercise 1.11.**

   Let $Z \sim N(\mu, 1)$, where $\mu \in \mathbb{R}$.

   (a) For $\lambda \geq 0$, define the ridge estimator

   $$\widehat{\mu}_\lambda(z) = \text{argmin}_{t \in \mathbb{R}} \left\{ \frac{1}{2}(t - z)^2 + \frac{\lambda}{2} t^2 \right\}.$$

   Compute

   $$\text{MSE}_\lambda(\mu) = \mathbb{E}\{(\widehat{\mu}_\lambda(Z) - \mu)^2\}.$$

   Plot and compare the risk functions for $\lambda = 0.1, 1, 5, 10$.

   (b) Repeat for the Lasso estimator

   $$\widehat{\mu}_\lambda(z) = \text{argmin}_{t \in \mathbb{R}} \left\{ \frac{1}{2}(t - z)^2 + \lambda |t| \right\}.$$

**Solution:**

**(a) Ridge.** The derivative of the objective is

$$(t - z) + \lambda t = (1 + \lambda)t - z.$$

Its second derivative is $1 + \lambda > 0$, so the unique minimizer is

$$\boxed{\widehat{\mu}_\lambda(z) = \frac{z}{1 + \lambda}.}$$

Since $\mathbb{E}Z = \mu$ and $\text{Var}(Z) = 1$,

$$\mathbb{E}\widehat{\mu}_\lambda(Z) - \mu = -\frac{\lambda \mu}{1 + \lambda}, \qquad \text{Var}\{\widehat{\mu}_\lambda(Z)\} = \frac{1}{(1 + \lambda)^2}.$$

Therefore

$$\boxed{\mathrm{MSE}_\lambda^{\mathrm{ridge}}(\mu) = \frac{1+\lambda^2\mu^2}{(1+\lambda)^2}.}$$

Each risk curve is even in $\mu$ and has its minimum at zero. Larger penalties reduce variance near zero but increase shrinkage bias for sufficiently large signals.

The unpenalized estimator $Z$ has MSE 1. For $\lambda > 0$,

$$\mathrm{MSE}_\lambda^{\mathrm{ridge}}(\mu) < 1 \iff 1+\lambda^2\mu^2 < (1+\lambda)^2 \iff \boxed{\mu^2 < 1 + \frac{2}{\lambda}.}$$

Thus a positive ridge penalty improves on $Z$ exactly on a bounded interval of signal sizes.

At $\mu = 0$, the risks for the four penalties are approximately

$$0.826446, \qquad 0.25, \qquad 0.027778, \qquad 0.008264.$$

**(b) Lasso.** The objective is the sum of the strictly convex function $\frac{1}{2}(t-z)^2$ and the convex function $\lambda|t|$, so it is strictly convex and has a unique minimizer, characterized by

$$0 \in t - z + \lambda \partial |t|.$$

Considering positive, zero, and negative $t$ gives

$$\boxed{\widehat{\mu}_\lambda(z) = S_\lambda(z) = \begin{cases} z-\lambda, & z > \lambda, \\ 0, & |z| \le \lambda, \\ z+\lambda, & z < -\lambda. \end{cases}}$$

Write $Z = \mu + U$, where $U \sim N(0,1)$, and define

$$a = -\lambda - \mu, \qquad b = \lambda - \mu.$$

Then

$$\widehat{\mu}_\lambda(Z) - \mu = \begin{cases} U+\lambda, & U < a, \\ -\mu, & a \le U \le b, \\ U-\lambda, & U > b. \end{cases}$$

Let $\phi$ and $\Phi$ denote the standard normal density and CDF. The MSE is

$$\begin{aligned} \mathrm{MSE}_\lambda^{\mathrm{lasso}}(\mu) = & \int_{-\infty}^{a} (u+\lambda)^2 \phi(u)\, du \\ & + \mu^2\{\Phi(b) - \Phi(a)\} \\ & + \int_b^{\infty} (u-\lambda)^2 \phi(u)\, du. \end{aligned}$$

Using $\phi'(u) = -u\phi(u)$ and integration by parts,

$$\begin{aligned}
\int_{-\infty}^{a} u\phi(u)\,du &= -\phi(a),\\
\int_{-\infty}^{a} u^2\phi(u)\,du &= \Phi(a) - a\phi(a),\\
\int_{b}^{\infty} u\phi(u)\,du &= \phi(b),\\
\int_{b}^{\infty} u^2\phi(u)\,du &= 1 - \Phi(b) + b\phi(b).
\end{aligned}$$

Expanding the squares and substituting yields

$$\boxed{\begin{aligned}
\mathrm{MSE}_\lambda^{\mathrm{lasso}}(\mu) = (1+\lambda^2)\{&\Phi(a) + 1 - \Phi(b)\}\\
&- (a + 2\lambda)\phi(a)\\
&+ (b - 2\lambda)\phi(b)\\
&+ \mu^2\{\Phi(b) - \Phi(a)\}.
\end{aligned}}$$

At $\lambda = 0$, the formula reduces to 1. At $\mu = 0$, it becomes

$$\mathrm{MSE}_\lambda^{\mathrm{lasso}}(0) = 2\{(1+\lambda^2)\Phi(-\lambda) - \lambda\phi(\lambda)\}.$$

The risk is even in $\mu$. Large penalties produce small risk near zero by setting the estimator to zero with high probability. For fixed $\lambda$,

$$\boxed{\lim_{|\mu|\to\infty} \mathrm{MSE}_\lambda^{\mathrm{lasso}}(\mu) = 1 + \lambda^2.}$$

To justify this, note $|S_\lambda(z) - z| \le \lambda$ for all $z$, so

$$\{S_\lambda(\mu + U) - \mu\}^2 \le (|U| + \lambda)^2,$$

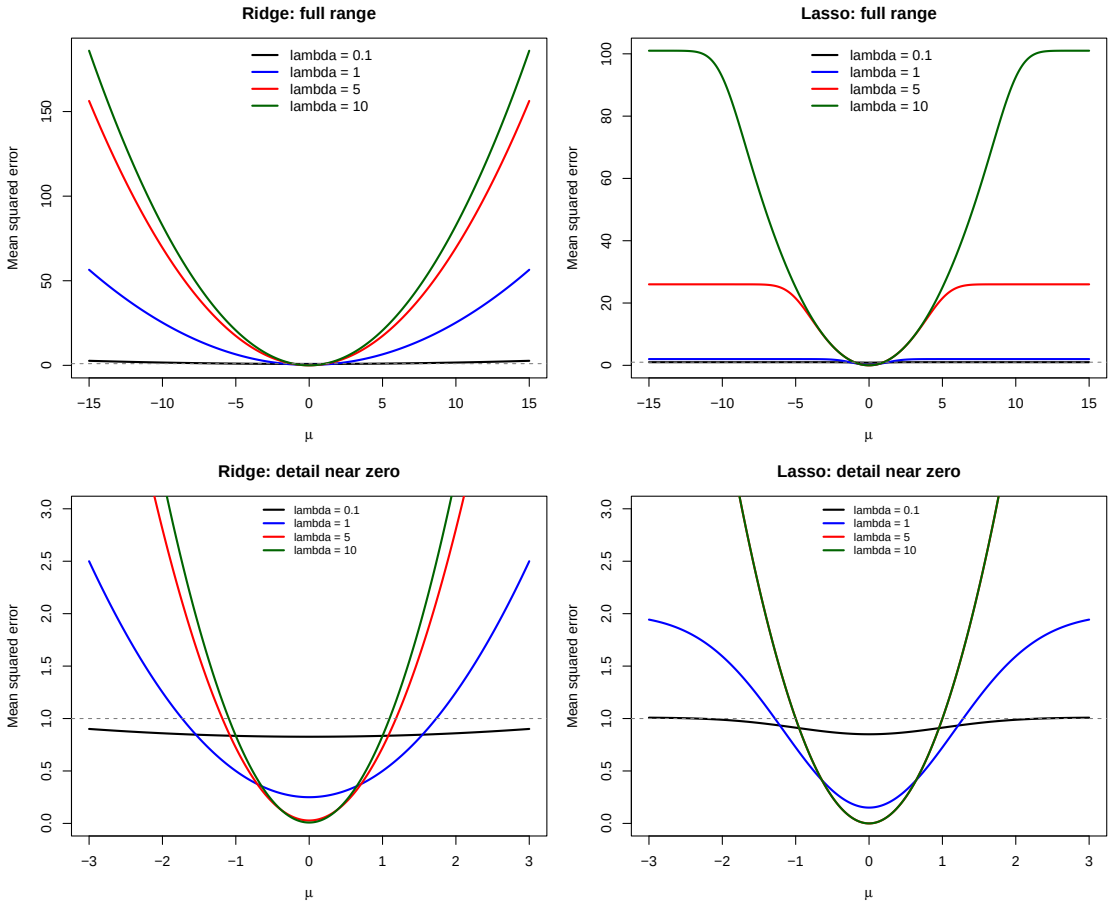
which is integrable and does not depend on $\mu$. For each fixed $U$, $\mu + U > \lambda$ once $\mu$ is large, so $S_\lambda(\mu + U) - \mu \to U - \lambda$ pointwise as $\mu \to \infty$. Dominated convergence gives $\mathrm{MSE}_\lambda^{\mathrm{lasso}}(\mu) \to \mathbb{E}(U - \lambda)^2 = 1 + \lambda^2$; evenness handles $\mu \to -\infty$. The limiting risks for the four penalties are

$$1.01, \qquad 2, \qquad 26, \qquad 101.$$

Unlike ridge, the Lasso risk approaches a finite plateau for each fixed penalty.

**Comparison.** Both estimators reduce risk near zero by shrinking toward it. Ridge shrinks proportionally, so for $\lambda > 0$ its bias $-\lambda\mu/(1+\lambda)$ grows linearly in $|\mu|$ and its

risk diverges as $|\mu| \to \infty$. Lasso subtracts at most $\lambda$, so its risk tends to the finite limit $1+\lambda^2$. Thus, for each fixed positive penalty, Lasso avoids the unbounded large-signal risk of ridge. At $\mu = 0$, ridge has the smaller risk for $\lambda = 0.1$ (0.826 versus 0.850), but Lasso has the smaller risk for $\lambda = 1, 5, 10$, because it sets the estimate exactly to zero with high probability. Both improve on the unpenalized estimator in a neighborhood of zero, but for $\lambda > 0$ neither dominates it uniformly: each has risk exceeding 1 for all sufficiently large $|\mu|$.




Exact ridge and Lasso MSE curves for $\lambda = 0.1, 1, 5, 10$: full range (top) and detail for $|\mu| \leq 3$ (bottom); the detail panels restrict the vertical range to emphasize differences near the unpenalized benchmark, so some curves are clipped above 3. The dashed line is the unpenalized risk 1.

2. **Exercise 1.12.**

Let $Y_i$ be responses and let $X_i = X_{i,1:p}$ denote covariates excluding a constant. Define

$$\widetilde{Y}_i = Y_i - \overline{Y}, \qquad \widetilde{X}_i = X_i - \overline{X}.$$

(a) For ridge regression with an unpenalized intercept, show that the slopes can be obtained from the centered data and that

$$\widehat{\beta}_0 = \overline{Y} - \overline{X}^{\mathsf{T}} \widetilde{\beta}_{1:p}.$$

Also show that

$$\operatorname{tr}(S_\lambda) = 1 + \operatorname{tr}(\widetilde{S}_\lambda).$$

(b) Prove the same centering and intercept identities for Lasso.

**Solution:**

**(a)** Write $b = \beta_{1:p}$. The ridge criterion is

$$Q(\beta_0, b) = \frac{1}{2n} \sum_{i=1}^{n} (Y_i - \beta_0 - X_i^{\mathsf{T}} b)^2 + \frac{\lambda}{2} \|b\|_2^2 .$$

For fixed $b$, the penalty is independent of the intercept. Differentiating in $\beta_0$ gives

$$\frac{\partial Q}{\partial \beta_0} = -\frac{1}{n} \sum_{i=1}^{n} (Y_i - \beta_0 - X_i^{\mathsf{T}} b).$$

The second derivative is 1, so the unique minimizing intercept is

$$\beta_0(b) = \overline{Y} - \overline{X}^{\mathsf{T}} b.$$

Substituting into the residual gives

$$Y_i - \beta_0(b) - X_i^{\mathsf{T}} b = \widetilde{Y}_i - \widetilde{X}_i^{\mathsf{T}} b.$$

Consequently the profiled objective is

$$\min_{\beta_0} Q(\beta_0, b) = \frac{1}{2n} \sum_{i=1}^{n} (\widetilde{Y}_i - \widetilde{X}_i^{\mathsf{T}} b)^2 + \frac{\lambda}{2} \|b\|_2^2 .$$

This is exactly the centered ridge problem. Therefore

$$\boxed{\widehat{\beta}_{1:p} = \widetilde{\beta}_{1:p}, \qquad \widehat{\beta}_0 = \overline{Y} - \overline{X}^{\mathsf{T}} \widetilde{\beta}_{1:p}.}$$

More precisely, if $\widetilde{b}$ is any minimizer of the centered problem, then $(\overline{Y} - \overline{X}^{\mathsf{T}} \widetilde{b}, \widetilde{b})$ minimizes the original problem; conversely, every minimizer of the original problem has this form, since its intercept must equal $\beta_0(b)$. Thus the sets of slope minimizers coincide, and the displayed equality is literal whenever the slope solution is unique (for example, for every $\lambda > 0$).

For the smoother identity, let $X$ be the original (uncentered) $n \times p$ covariate matrix and define

$$J = \frac{1}{n}\mathbf{1}\mathbf{1}^\mathsf{T}, \qquad H = I - J.$$

Then

$$\widetilde{X} = HX, \qquad \widetilde{Y} = HY.$$

For $\lambda > 0$, the centered smoother is

$$\widetilde{S}_\lambda = \widetilde{X}(\widetilde{X}^\mathsf{T}\widetilde{X} + n\lambda I)^{-1}\widetilde{X}^\mathsf{T}.$$

Because $\widetilde{X}^\mathsf{T}\mathbf{1} = 0$,

$$\widetilde{S}_\lambda J = 0, \qquad \widetilde{S}_\lambda H = \widetilde{S}_\lambda.$$

The fitted response from the original regression is

$$\begin{aligned}\widehat{Y} &= \mathbf{1}\widehat{\beta}_0 + X\widetilde{\beta}_{1:p} \\ &= \overline{Y}\mathbf{1} + \widetilde{X}\widetilde{\beta}_{1:p} \\ &= JY + \widetilde{S}_\lambda HY \\ &= (J + \widetilde{S}_\lambda)Y.\end{aligned}$$

Thus

$$S_\lambda = J + \widetilde{S}_\lambda.$$

Since $\operatorname{tr}(J) = 1$,

$$\boxed{\operatorname{tr}(S_\lambda) = 1 + \operatorname{tr}(\widetilde{S}_\lambda).}$$

At $\lambda = 0$, use the centered least-squares projection matrix, defined with a Moore–Penrose inverse when necessary. The same fitted-value argument applies.

**(b)** For Lasso, the objective becomes

$$Q(\beta_0, b) = \frac{1}{2n}\sum_{i=1}^{n}(Y_i - \beta_0 - X_i^\mathsf{T} b)^2 + \lambda\|b\|_1.$$

The penalty still does not involve $\beta_0$, so the optimal intercept and residual substitution are unchanged, and the same correspondence of minimizer sets holds. Hence

$$\boxed{\widehat{\beta}_{1:p} = \widetilde{\beta}_{1:p}, \qquad \widehat{\beta}_0 = \overline{Y} - \overline{X}^\mathsf{T}\widetilde{\beta}_{1:p}.}$$

No differentiation of the Lasso penalty is needed.

3. **Exercise 1.13.**

Let

$$s_j = \|X_j\|_n = \left(\frac{1}{n}\sum_{i=1}^{n} X_{ij}^2\right)^{1/2}, \qquad \widetilde{X}_{ij} = \frac{X_{ij}}{s_j}.$$

(a) Find weights $c_j$ so that fitting an unweighted ridge penalty to the standardized covariates and setting $\widehat{\beta}_j = \widehat{\gamma}_j / s_j$ is equivalent to fitting a weighted ridge penalty to the original covariates.

(b) Find the corresponding weights for Lasso.

**Solution:**

Assume $s_j > 0$ for each $j$, since otherwise the stated standardization is undefined.

Let

$$D = \mathrm{diag}(s_1, \ldots, s_p).$$

Then $\widetilde{X} = XD^{-1}$, and the change of variables

$$\gamma = D\beta$$

is a bijection satisfying

$$\widetilde{X}\gamma = X\beta.$$

**(a) Ridge.** The standardized objective becomes

$$\begin{aligned}\frac{1}{2n}\|Y - \widetilde{X}\gamma\|_2^2 + \frac{\lambda}{2}\sum_{j=1}^{p}\gamma_j^2 \\ = \frac{1}{2n}\|Y - X\beta\|_2^2 + \frac{\lambda}{2}\sum_{j=1}^{p} s_j^2\beta_j^2.\end{aligned}$$

The objectives agree under the bijection, so their minimizers correspond. Therefore

$$\boxed{c_j = s_j^2 = \frac{1}{n}\sum_{i=1}^{n} X_{ij}^2.}$$

**(b) Lasso.** Similarly,

$$\begin{aligned}\frac{1}{2n}\|Y - \widetilde{X}\gamma\|_2^2 + \lambda\sum_{j=1}^{p}|\gamma_j| \\ = \frac{1}{2n}\|Y - X\beta\|_2^2 + \lambda\sum_{j=1}^{p} s_j|\beta_j|.\end{aligned}$$

Thus

$$\boxed{c_j = s_j = \left(\frac{1}{n}\sum_{i=1}^{n} X_{ij}^2\right)^{1/2}.}$$

Standardization therefore changes the relative penalty weights on the original coefficients: a predictor with larger empirical scale $s_j$ receives a larger penalty on its original-scale coefficient, quadratically in $s_j$ for ridge and linearly for Lasso. These weights make the original-scale problem equivalent to penalizing all standardized coefficients equally. Here $s_j$ is the uncentered root mean square defined above, not the centered sample standard deviation.

4. **Exercise 1.15.**

   (a) For the Lasso objective

   $$h(\beta) = \frac{1}{2n}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1,$$

   show that coordinate-wise minimization implies joint minimization.

   (b) Generalize to

   $$h(\beta) = h_0(\beta) + \sum_{j=1}^{p} h_{1,j}(\beta_j),$$

   where $h_0$ is convex and everywhere differentiable, and each $h_{1,j}$ is convex but may be nondifferentiable.

   (c) Give a convex counterexample for which coordinate-wise minimization does not imply joint minimization.

**Solution:**

**(a)** Let

$$h_0(\beta) = \frac{1}{2n}\|Y - X\beta\|_2^2.$$

Its gradient is

$$\nabla h_0(\beta) = -\frac{1}{n}X^\top(Y - X\beta).$$

Suppose $\widetilde{\beta}$ minimizes the objective in every coordinate when the other coordinates are held fixed. For each $j$ define the one-dimensional convex function

$$g_j(t) = h_0(\widetilde{\beta}_1, \ldots, \widetilde{\beta}_{j-1}, t, \widetilde{\beta}_{j+1}, \ldots, \widetilde{\beta}_p) + \lambda|t| + \lambda\sum_{k\neq j}|\widetilde{\beta}_k|.$$

Coordinate-wise optimality means $\widetilde{\beta}_j$ minimizes $g_j$, so $0 \in \partial g_j(\widetilde{\beta}_j)$. Since the first term is differentiable, the subdifferential sum rule gives the one-dimensional optimality condition

$$0 \in \frac{\partial h_0}{\partial \beta_j}(\widetilde{\beta}) + \lambda \partial |\widetilde{\beta}_j|.$$

The absolute-value subdifferential is

$$\partial |t| = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0. \end{cases}$$

For each $j$, select $s_j \in \partial |\widetilde{\beta}_j|$ such that

$$\frac{\partial h_0}{\partial \beta_j}(\widetilde{\beta}) + \lambda s_j = 0.$$

Since the $L_1$ penalty is separable,

$$s = (s_1, \ldots, s_p)^{\mathsf{T}} \in \partial \|\widetilde{\beta}\|_1.$$

Combining the coordinate conditions gives

$$0 = \nabla h_0(\widetilde{\beta}) + \lambda s \in \partial h(\widetilde{\beta}).$$

By the subgradient inequality, for every $\beta$,

$$h(\beta) \geq h(\widetilde{\beta}) + 0^{\mathsf{T}}(\beta - \widetilde{\beta}) = h(\widetilde{\beta}).$$

Therefore $\widetilde{\beta}$ is a global minimizer.

This establishes the optimality of a coordinate-wise minimizing point; it is not by itself a proof of convergence of every coordinate-descent implementation.

**(b)** Assume each $h_{1,j}$ is a finite-valued convex function on $\mathbb{R}$, so that $\partial h_{1,j}(t)$ is nonempty for every $t$ and the subdifferential sum rule applies to $g_j$ below. As in part (a), apply $0 \in \partial g_j(\widetilde{\beta}_j)$ to $g_j(t) = h_0(\widetilde{\beta}_1, \ldots, t, \ldots, \widetilde{\beta}_p) + h_{1,j}(t)$ (plus constants); because $h_0$ is differentiable, $\partial g_j(\widetilde{\beta}_j) = \partial_j h_0(\widetilde{\beta}) + \partial h_{1,j}(\widetilde{\beta}_j)$. Hence coordinate-wise optimality implies that, for each $j$, there exists

$$s_j \in \partial h_{1,j}(\widetilde{\beta}_j)$$

satisfying

$$s_j = -\frac{\partial h_0}{\partial \beta_j}(\widetilde{\beta}).$$

Convexity of $h_0$ gives

$$h_0(\beta) \geq h_0(\widetilde{\beta}) + \nabla h_0(\widetilde{\beta})^{\mathsf{T}}(\beta - \widetilde{\beta}).$$

The subgradient inequality for each penalty term gives

$$h_{1,j}(\beta_j) \geq h_{1,j}(\widetilde{\beta}_j) + s_j(\beta_j - \widetilde{\beta}_j).$$

Adding these inequalities,

$$\begin{aligned} h(\beta) &\geq h(\widetilde{\beta}) + \sum_{j=1}^{p} \left\{ \frac{\partial h_0}{\partial \beta_j}(\widetilde{\beta}) + s_j \right\} (\beta_j - \widetilde{\beta}_j) \\ &= h(\widetilde{\beta}). \end{aligned}$$

Hence $\widetilde{\beta}$ is jointly minimizing. Separability allows the individual penalty subgradients to be chosen simultaneously.

**(c)** Consider

$$\boxed{h(x, y) = |x - y| + \frac{1}{4}(x + y - 2)^2.}$$

The function is convex because it is the sum of the absolute value of a linear function and the square of an affine function.

At $(0, 0)$, fixing $y = 0$ gives

$$h(x, 0) - h(0, 0) = |x| - x + \frac{x^2}{4} \geq 0.$$

Equality holds only at $x = 0$. By symmetry,

$$h(0, y) - h(0, 0) = |y| - y + \frac{y^2}{4} \geq 0,$$

with equality only at $y = 0$.

Thus $(0, 0)$ uniquely minimizes the objective along each coordinate line. However,

$$h(0, 0) = 1, \qquad h(1, 1) = 0.$$

Since $h \geq 0$, $(1, 1)$ is a global minimizer and $(0, 0)$ is not.

The nondifferentiable term $|x - y|$ couples the coordinates; it does not satisfy the separability assumption in part (b).

5. **Exercise 1.17.**

Using the provided simulation code (seed 0), generate $n = 200$ training and $n_{\text{test}} = 100$ test observations with $p = 20$,

$$X \sim N_p(0, \Sigma), \qquad \Sigma_{jk} = 0.8^{|j-k|},$$

$$Y = X_1 + \tfrac{1}{3}\{\max(0, X_2 + X_3 + X_4)\}^2 + \varepsilon, \qquad \varepsilon \sim N(0, 0.5^2).$$

Apply all methods without standardizing $x$ or $y$.

(a) Fit ridge regression with an unpenalized intercept and produce a coefficient path.

(b) Plot training, test, 5-fold CV, and 10-fold CV errors against degrees of freedom. Report the minimum test error and test errors at the minimum-CV and one-standard-error choices.

(c) Fit Lasso with an unpenalized intercept and produce a coefficient path.

(d) Plot the four errors against the shrinkage fraction and report the corresponding test-error comparisons.

(e) Summarize the findings in one paragraph.

**Solution:**

**Setup.** Write $m(X) = X_1 + \frac{1}{3}\{\max(0, X_2 + X_3 + X_4)\}^2$ for the generating conditional mean. All computations are done by `hw2.R`, submitted separately (see the reproducibility note at the end). The script generates both samples before drawing CV folds (fold seed 123), and ridge and Lasso use the same folds. Neither $x$ nor $y$ is rescaled; centering profiles out the unpenalized intercept (Exercise 1.12).

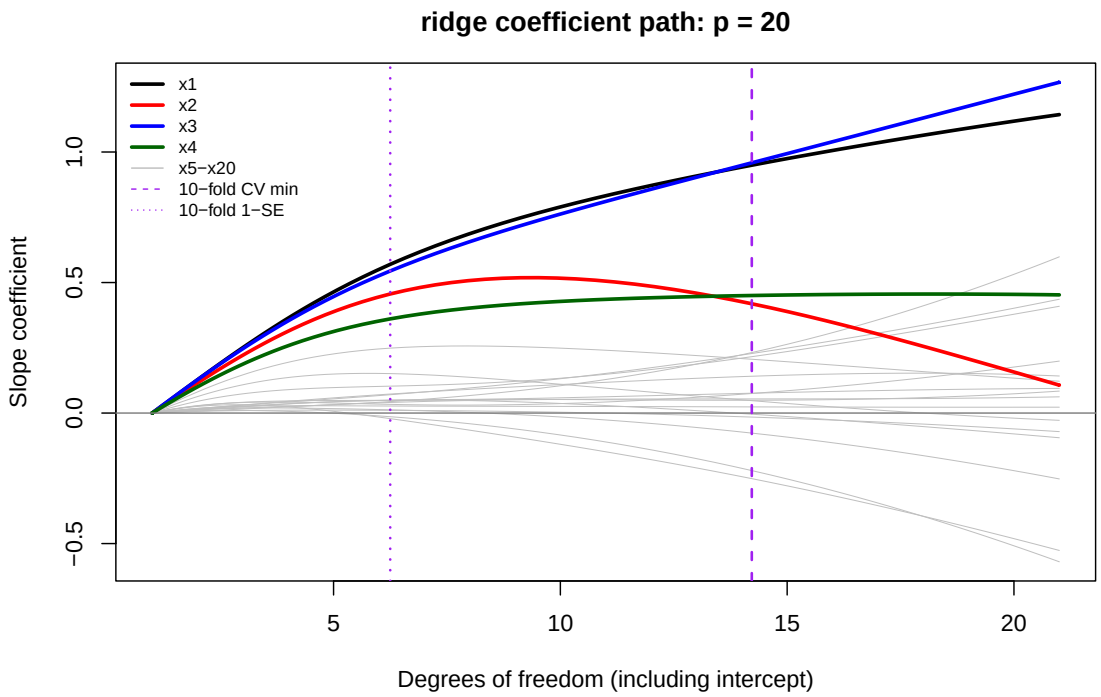
**(a) Ridge.** For each penalty,

$$\widehat{b}_\lambda = (\widetilde{X}^\mathsf{T}\widetilde{X} + n\lambda I)^{-1}\widetilde{X}^\mathsf{T}\widetilde{Y}, \qquad \widehat{\beta}_{0,\lambda} = \overline{Y} - \overline{X}^\mathsf{T}\widehat{b}_\lambda,$$

computed via an SVD of $\widetilde{X}$. If $d_1, \ldots, d_r$ are the nonzero singular values of $\widetilde{X}$, the effective degrees of freedom are

$$\boxed{\mathrm{df}(\lambda) = 1 + \sum_{j=1}^{r} \frac{d_j^2}{d_j^2 + n\lambda},}$$

where the intercept contributes the leading 1. The grid contains 240 positive penalties from $10^4 s$ to $10^{-8} s$, where $s = d_{\max}^2/n$.




Ridge coefficient path versus effective degrees of freedom. The four predictors in the generating mean are highlighted; the others are gray. Vertical lines mark the 10-fold CV choices.

**(b) Errors and cross-validation.** Training and test errors are mean squared prediction errors:

$$\mathrm{Err}_{\mathrm{train}}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{Y}_{i,\lambda})^2, \qquad \mathrm{Err}_{\mathrm{test}}(\lambda) = \frac{1}{n_{\mathrm{test}}}\sum_{i=1}^{n_{\mathrm{test}}}(Y_i^{\mathrm{test}} - \widehat{Y}_{i,\lambda}^{\mathrm{test}})^2.$$

For $K \in \{5, 10\}$, let $e_k(\lambda)$ be the mean squared held-out error in fold $k$, using a fit trained on the other folds. The folds have equal size, so

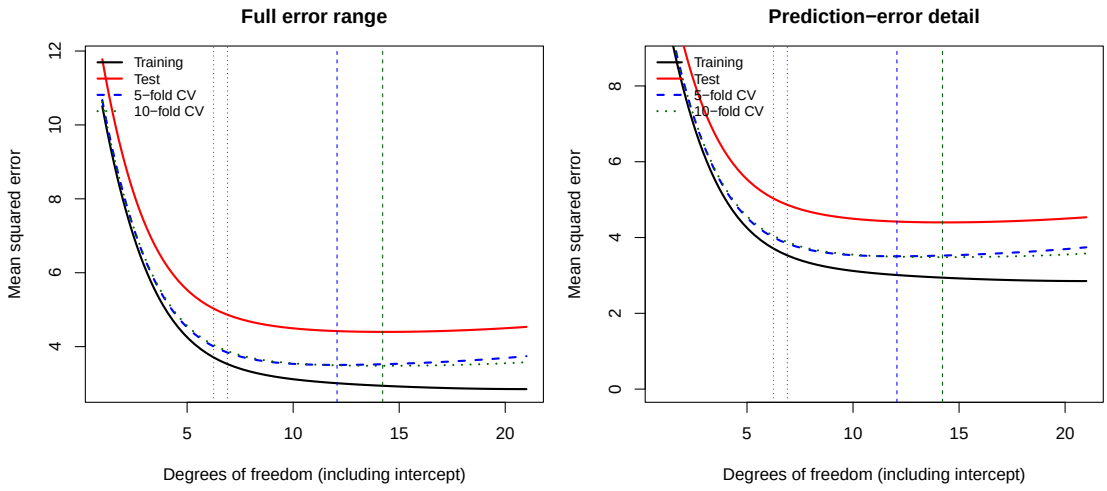
$$\mathrm{CV}_K(\lambda) = \frac{1}{K}\sum_{k=1}^{K} e_k(\lambda), \qquad \mathrm{SE}_K(\lambda) = \frac{\mathrm{sd}\{e_1(\lambda), \ldots, e_K(\lambda)\}}{\sqrt{K}}.$$

The minimum-CV choice is $\lambda_{\min,K} \in \operatorname{argmin}_{\lambda\in\Lambda} \mathrm{CV}_K(\lambda)$. The one-standard-error choice is the largest penalty satisfying

$$\mathrm{CV}_K(\lambda) \le \mathrm{CV}_K(\lambda_{\min,K}) + \mathrm{SE}_K(\lambda_{\min,K}),$$

so the one-SE rule favors stronger regularization among fits whose CV error is close to the minimum. This standard error is a fold-based selection heuristic; overlapping training sets do not produce independent fits.

CV is computed at common penalty values across folds. For plotting, each penalty is mapped to the degrees of freedom (ridge) or shrinkage fraction (Lasso) of its full-training-data fit.




Ridge training, test, and CV errors. The plots show the full error range and a detailed view.

| Selection | $\lambda$ | df | Test MSE |
|---|---|---|---|
| Test-grid oracle | 0.1328 | 14.2 | 4.3991 |
| 5-fold CV minimum | 0.2367 | 12.1 | 4.4204 |
| 5-fold CV 1-SE | 1.064 | 6.9 | 4.8629 |
| 10-fold CV minimum | 0.1328 | 14.2 | 4.3991 |
| 10-fold CV 1-SE | 1.341 | 6.25 | 5.0294 |

The df column locates each selected fit on the horizontal axis. The test-grid oracle is the smallest observed test error among the 240 fitted penalties. It is reported only as a benchmark: test data are not used to select the CV penalties, and a minimum over a finite grid need not equal the minimum over all $\lambda \geq 0$.

**(c) Lasso.** The objective is

$$\frac{1}{2n}\|Y - \beta_0\mathbf{1} - Xb\|_2^2 + \lambda\|b\|_1,$$

fitted with `glmnet` (`alpha = 1`, `standardize = FALSE`, `intercept = TRUE`). Define

$$\lambda_{\max} = \max_j \left|\frac{1}{n}\widetilde{X}_j^{\mathsf{T}}\widetilde{Y}\right|,$$

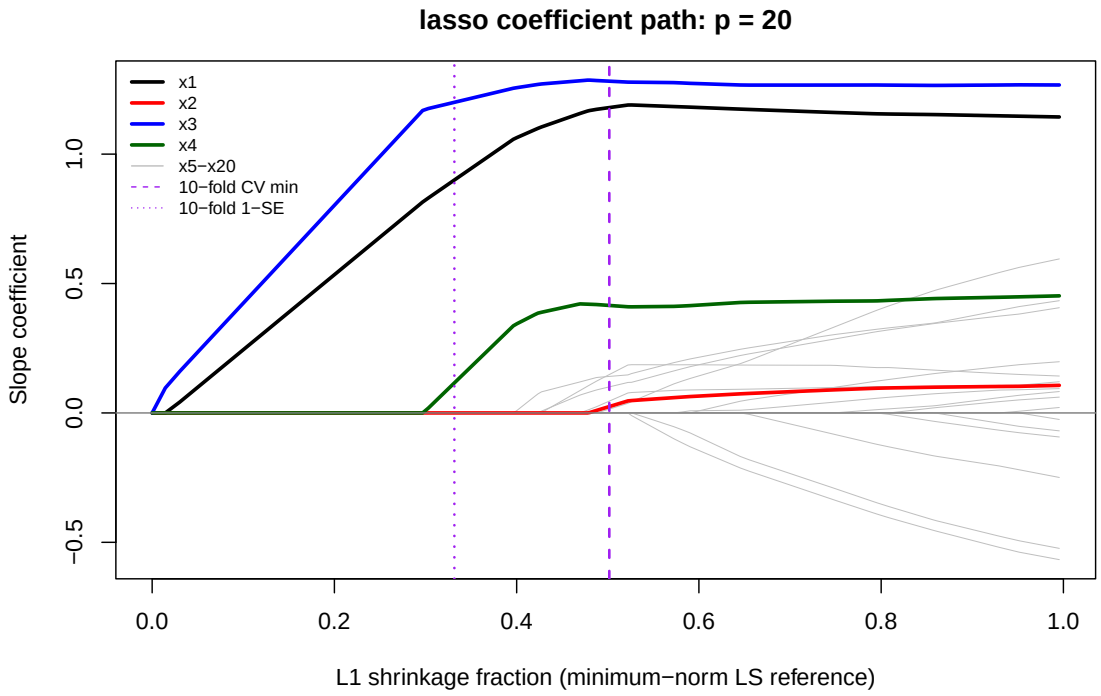
the smallest penalty at which zero slopes satisfy the stationary conditions. The grid has 240 log-spaced penalties from $\lambda_{\max}$ to $10^{-4}\lambda_{\max}$.

For the full-data path and every CV training fit, the script checks the original-scale KKT conditions, including the zero-mean-residual condition for the intercept, and requires the maximum violation to be below $10^{-6}\lambda_{\max}$. These checks are strong evidence that the returned solutions are near-optimal, although small KKT residuals do not by themselves certify that every coefficient is accurate to the same tolerance.

The shrinkage fraction is

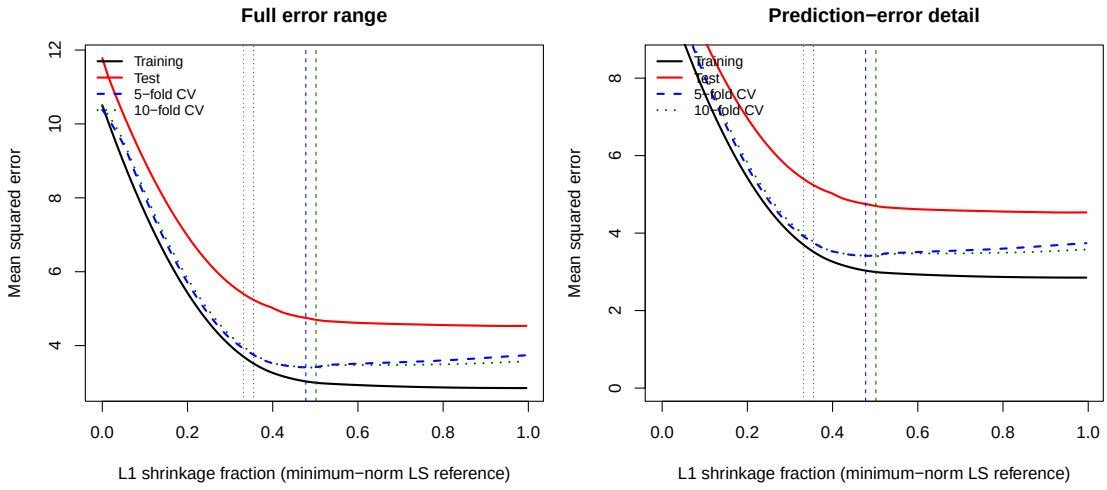
$$f(\lambda) = \frac{\|\widehat{b}_\lambda\|_1}{\|\widehat{b}_{\mathrm{LS}}\|_1}.$$

For $p = 20$, the centered design has full column rank almost surely, so the least-squares slope reference is unique, and $f(\lambda) \in [0, 1]$ is nonincreasing in $\lambda$ (proved in Exercise 1.18). Throughout, a slope is counted as active if $|\widehat{b}_j| > 10^{-8}$.




Lasso coefficient path versus the $L_1$ shrinkage fraction.

**(d) Lasso errors.** The error definitions, folds, and selection rules are those of part (b), now plotted against $f(\lambda)$.




Lasso training, test, and CV errors.

| Selection | $\lambda$ | $f(\lambda)$ | Nonzero | Kept of $x_1$–$x_4$ | Test MSE |
|---|---|---|---|---|---|
| Test-grid oracle | 0.002692 | 0.951 | 19 | 1,2,3,4 | 4.5319 |
| 5-fold CV minimum | 0.1372 | 0.478 | 7 | 1,3,4 | 4.7514 |
| 5-fold CV 1-SE | 0.5285 | 0.355 | 3 | 1,3,4 | 5.2372 |
| 10-fold CV minimum | 0.08978 | 0.502 | 9 | 1,2,3,4 | 4.6991 |
| 10-fold CV 1-SE | 0.6408 | 0.332 | 3 | 1,3,4 | 5.3977 |

The $f(\lambda)$ column locates each selected fit on the plots; "Nonzero" counts active slopes, and the next column lists which of $X_1, \dots, X_4$ are retained.

**Benchmark: the best linear predictor.** Both methods fit linear models, so the natural target is the best linear predictor $f^*(X) = \beta_0^* + X^\mathsf{T} b^*$ minimizing $\mathbb{E}(Y - \beta_0 - X^\mathsf{T} b)^2$, with $b^* = \Sigma^{-1} \operatorname{Cov}\{X, m(X)\}$. Since $X \sim N(0, \Sigma)$ and $m$ is continuously differentiable, Stein's lemma gives $\operatorname{Cov}\{X, m(X)\} = \Sigma\, \mathbb{E}\{\nabla m(X)\}$, so

$$b^* = \mathbb{E}\{\nabla m(X)\} = (1, c, c, c, 0, \dots, 0)^\mathsf{T}, \qquad c = \tfrac{2}{3}\mathbb{E}(S_+) = \tfrac{2}{3}\sqrt{v}\,\phi(0) \approx 0.727394,$$

where $S = X_2 + X_3 + X_4 \sim N(0, v)$ with $v = \operatorname{Var}(S) = 3 + 2(0.8 + 0.8 + 0.64) = 7.48$. The slopes on $X_5, \dots, X_p$ are exactly zero for every $p$, and the intercept is $\beta_0^* = \mathbb{E}\, g(S) = v/6$, where $g(S) = S_+^2/3$.

The residual of this predictor is $g(S) - v/6 - cS + \varepsilon$. Using $\mathbb{E}S_+^4 = 3v^2/2$ and $\mathbb{E}S_+^3 = \sqrt{2/\pi}\, v^{3/2}$, we get $\operatorname{Var}\{g(S)\} = 5v^2/36$ and $\operatorname{Cov}\{S, g(S)\} = cv$. Hence

$$R^*_{\text{linear}} = 0.25 + \operatorname{Var}\{g(S)\} - c^2 v = 0.25 + v^2\Big(\frac{5}{36} - \frac{2}{9\pi}\Big) \approx 4.0632.$$

This is the minimum population prediction risk among affine predictors (the *linear floor*): noise variance 0.25 plus linear approximation error 3.8132. (A Monte Carlo

check of this value is included in the script output.) For any affine fitted predictor $\widehat{f}$ trained on data $\mathcal{D}$, orthogonality of the best-linear-predictor residual to affine functions gives

$$R(\widehat{f} \mid \mathcal{D}) = \mathbb{E}\{(Y - \widehat{f}(X))^2 \mid \mathcal{D}\} = R^*_{\text{linear}} + \mathbb{E}\{(\widehat{f}(X) - f^*(X))^2 \mid \mathcal{D}\},$$

so regularization can affect only the second, excess-risk term. If $\widehat{f}$ is selected using training data only, the test sample is independent of $\mathcal{D}$, so

$$\mathbb{E}\big[\widehat{R}_{\text{test}} - R^*_{\text{linear}} \mid \mathcal{D}\big] = R(\widehat{f} \mid \mathcal{D}) - R^*_{\text{linear}}.$$

Thus test MSE minus the floor is an unbiased but noisy estimate of the conditional excess risk; it can be negative even though the excess risk itself cannot. This does not apply to the test-grid oracle, whose penalty is chosen using the same test outcomes.

**Paired comparison.** Squared test errors are highly skewed and variable, so a 100-observation test MSE can be noisy. The table compares ridge and Lasso on the same test points: for each rule it reports the mean of $(Y_i - \widehat{Y}_i^{\text{ridge}})^2 - (Y_i - \widehat{Y}_i^{\text{lasso}})^2$, its standard error over the 100 test points, and the ratio $z$. These standard errors describe test-sample uncertainty conditional on the training data and the selected models; they do not account for training-sample variability. The test-grid-oracle row is descriptive only.

| Selection | Ridge | Lasso | Ridge − Lasso | SE | z |
|---|---|---|---|---|---|
| Test-grid oracle | 4.399 | 4.532 | -0.133 | 0.125 | -1.059 |
| 5-fold CV minimum | 4.420 | 4.751 | -0.331 | 0.186 | -1.782 |
| 5-fold CV 1-SE | 4.863 | 5.237 | -0.374 | 0.265 | -1.415 |
| 10-fold CV minimum | 4.399 | 4.699 | -0.300 | 0.170 | -1.768 |
| 10-fold CV 1-SE | 5.029 | 5.398 | -0.368 | 0.278 | -1.326 |

**(e) Findings.** For $p = 20$, every tuned fit has test MSE between 4.40 and 5.40, against a linear floor of 4.06. Most of the population linear prediction risk is approximation error (3.81 of 4.06), which regularization cannot remove; regularization changes only the excess risk above the best linear predictor. For the CV-selected fits, the gap between test MSE and the floor estimates that excess risk without bias, but with substantial noise from the 100-point test sample. Minimum-CV tuning came close to the test-grid benchmark: 10-fold ridge selected the benchmark penalty exactly, and Lasso was within 0.17–0.22 of its benchmark. Ridge was lower than Lasso under every matching rule, by 0.13–0.37, but the paired standard errors are 0.13–0.28 ($|z| \leq 1.8$), so the paired differences provide limited evidence of a difference in conditional prediction risk on this realization. The one-SE rule chose penalties about 4–10 times larger and raised test MSE by 0.44–0.70 for both methods. The Lasso test-grid benchmark is close to the least-squares end of its path ($f \approx 0.95$), whereas the ridge

benchmark retains about 14.2 effective degrees of freedom out of a maximum of 21; stronger one-SE regularization worsened observed prediction performance for both methods in this realization. The one-SE Lasso fits keep only 3 slopes and drop $X_2$.

6. **Exercise 1.18.**

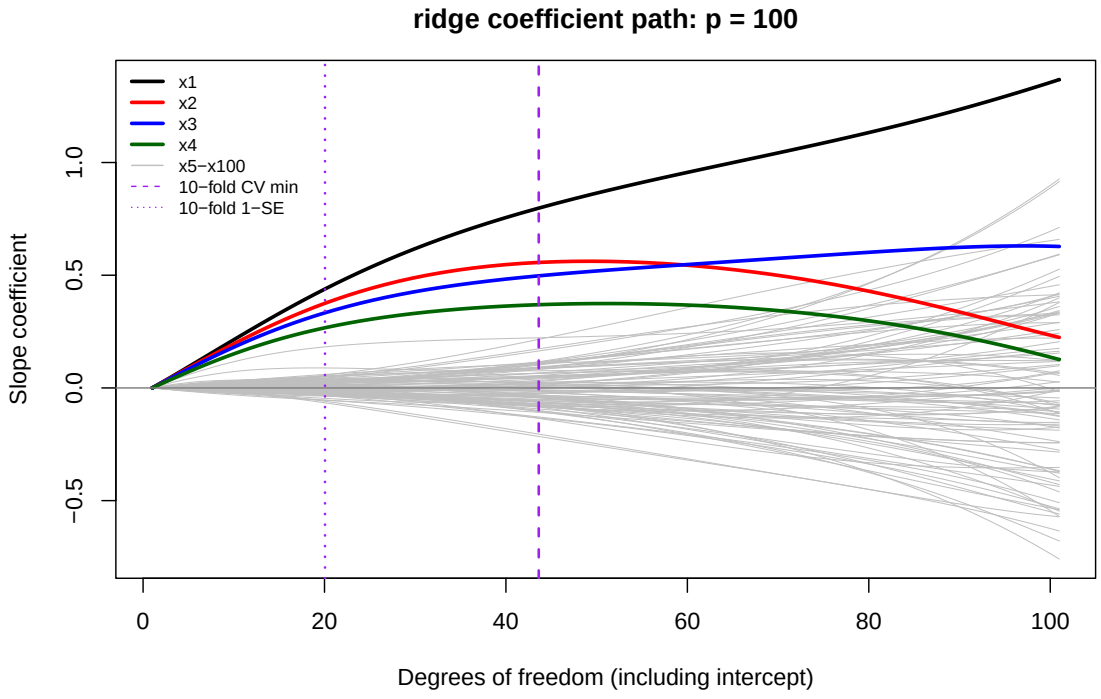
Repeat Exercise 1.17 with $p = 100$ and $p = 200$. For each case, answer parts (a)–(d) of Exercise 1.17, then summarize the findings in one paragraph.

**Solution:**

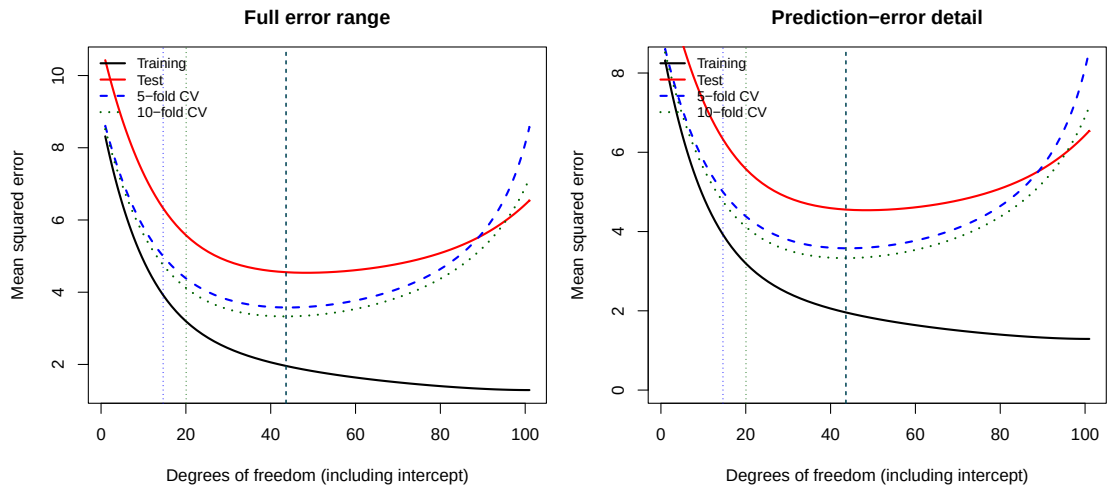
Each dimension uses the same simulation code as Exercise 1.17, with only $p$ changed. The seed is reset to zero before each complete training/test simulation, and the CV seed is applied afterward. The fitting methods, error definitions, plotting convention, and selection rules are unchanged.

**The $p = 100$ case.**

The ridge results are



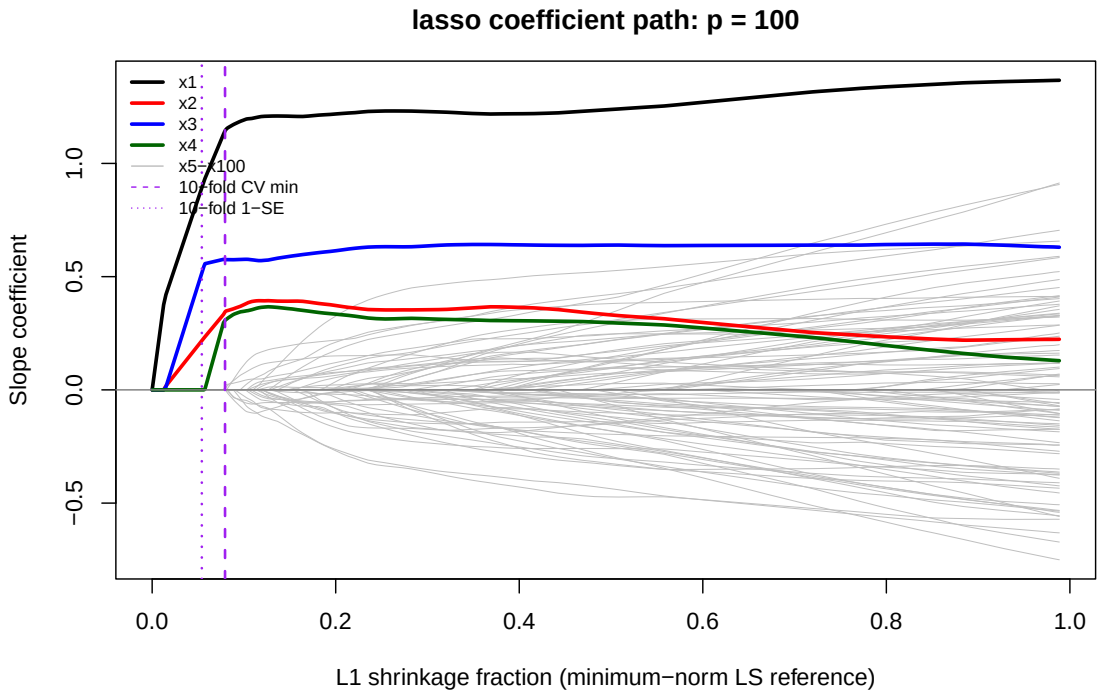

Ridge coefficient path, $p = 100$.



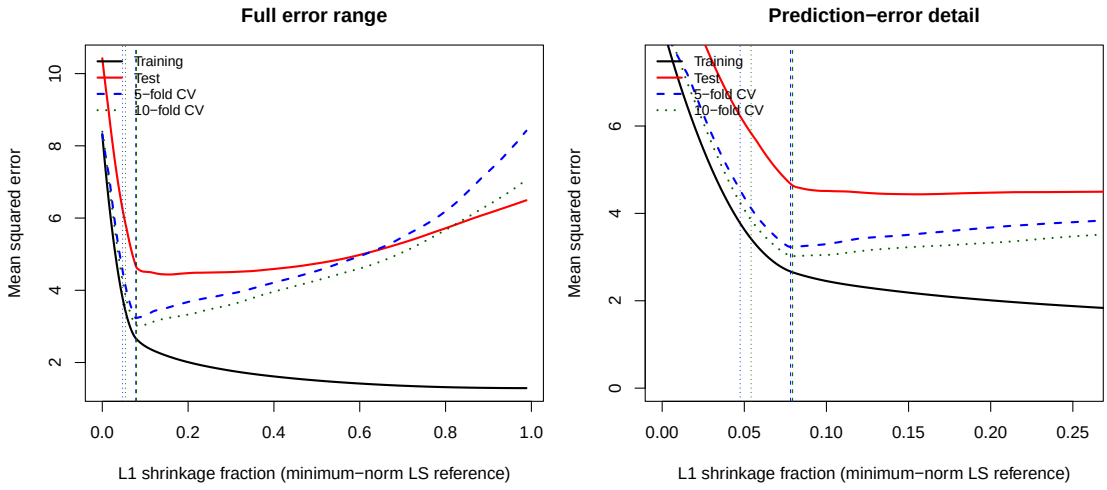

Ridge training, test, and CV errors, $p = 100$.

| Selection | $\lambda$ | df | Test MSE |
|---|---|---|---|
| Test-grid oracle | 0.2623 | 49.4 | 4.5384 |
| 5-fold CV minimum | 0.3711 | 43.6 | 4.5538 |
| 5-fold CV 1-SE | 3.747 | 14.6 | 6.3267 |
| 10-fold CV minimum | 0.3711 | 43.6 | 4.5538 |
| 10-fold CV 1-SE | 2.102 | 20.1 | 5.5740 |

The Lasso results are




Lasso coefficient path, $p = 100$.

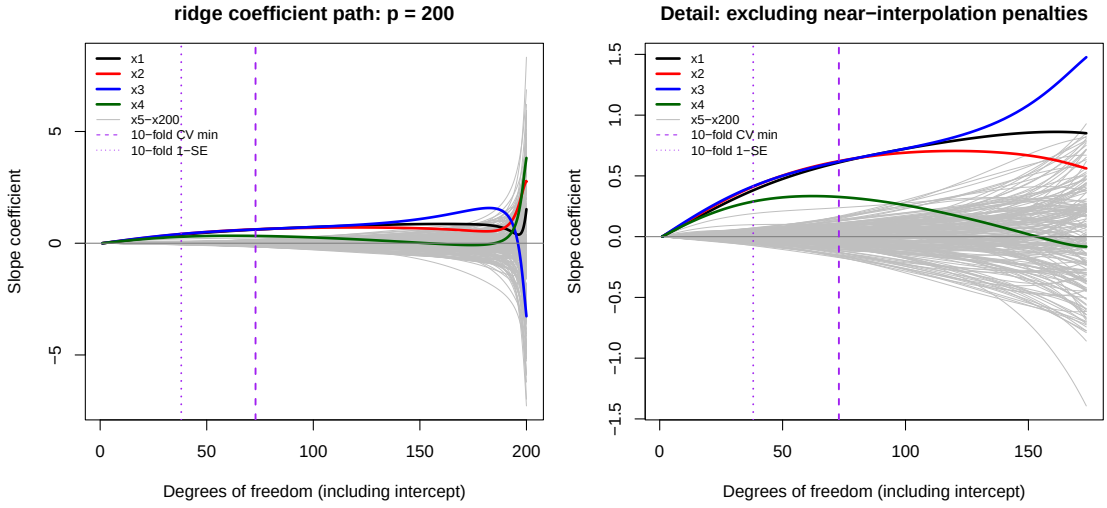



Lasso training, test, and CV errors, $p = 100$.

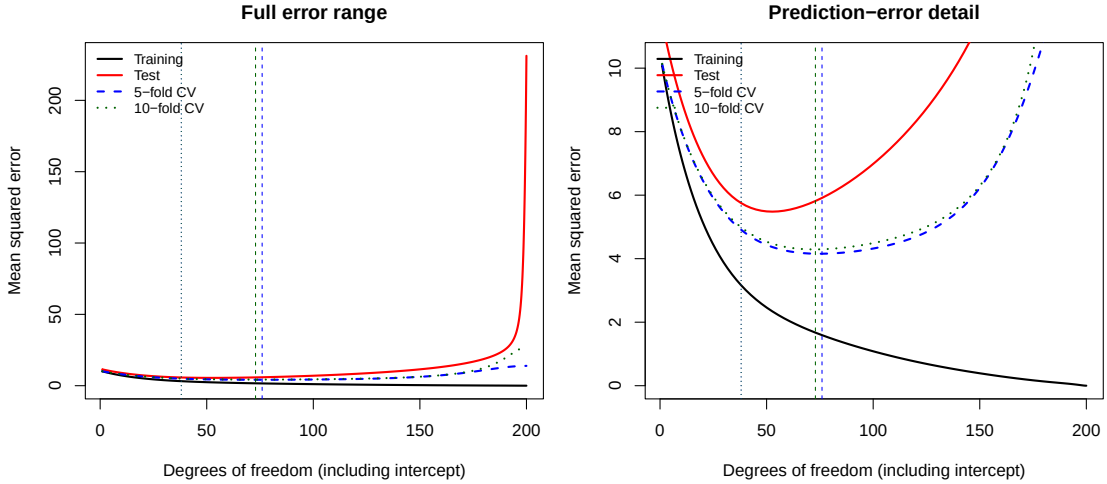
| Selection | $\lambda$ | $f(\lambda)$ | Nonzero | Kept of $x_1$–$x_4$ | Test MSE |
|---|---|---|---|---|---|
| Test-grid oracle | 0.06817 | 0.153 | 32 | 1,2,3,4 | 4.4370 |
| 5-fold CV minimum | 0.2432 | 0.0782 | 5 | 1,2,3,4 | 4.6753 |
| 5-fold CV 1-SE | 0.9736 | 0.0475 | 3 | 1,2,3 | 6.2272 |
| 10-fold CV minimum | 0.2166 | 0.0794 | 5 | 1,2,3,4 | 4.6373 |
| 10-fold CV 1-SE | 0.803 | 0.0541 | 3 | 1,2,3 | 5.8349 |

**The $p = 200$ case.**

The ridge results are



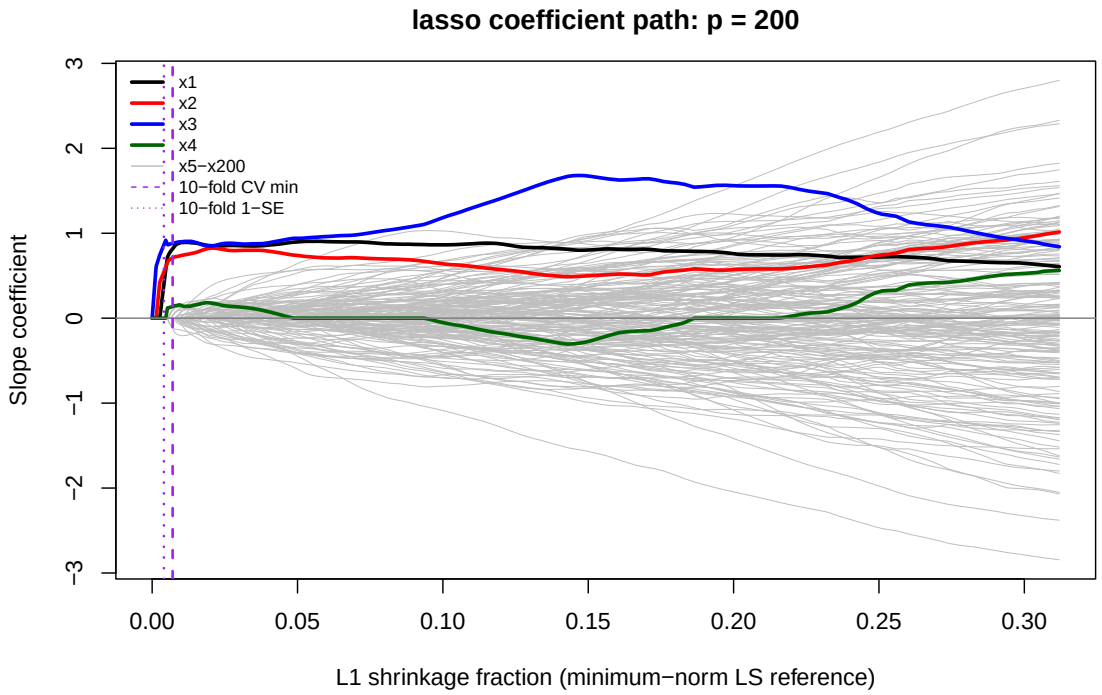

Ridge coefficient path, $p = 200$.



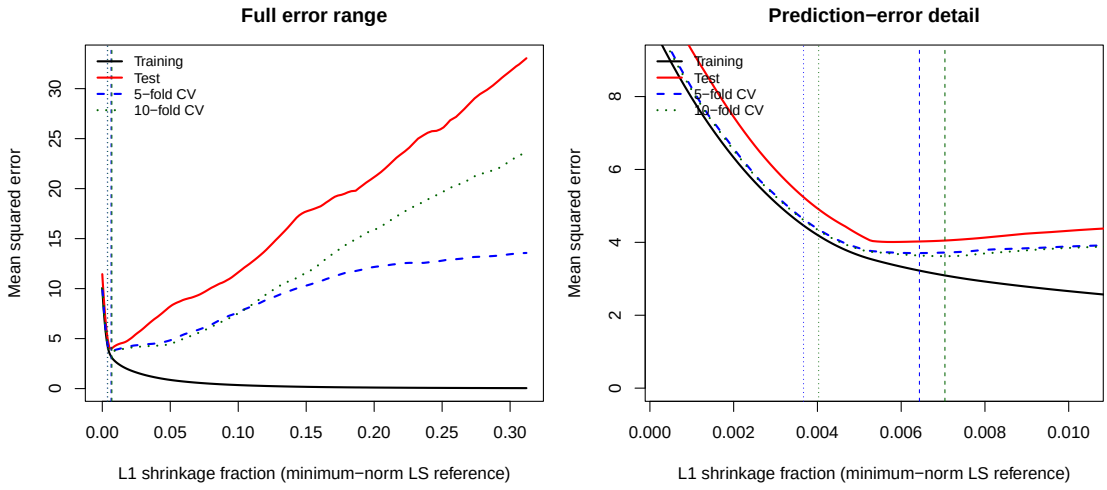

Ridge training, test, and CV errors, $p = 200$.

| Selection | $\lambda$ | df | Test MSE |
|---|---|---|---|
| Test-grid oracle | 0.9933 | 52.7 | 5.4811 |
| 5-fold CV minimum | 0.3939 | 76 | 5.9133 |
| 5-fold CV 1-SE | 1.988 | 38 | 5.7514 |
| 10-fold CV minimum | 0.4422 | 72.9 | 5.8159 |
| 10-fold CV 1-SE | 1.988 | 38 | 5.7514 |

The Lasso results are




Lasso coefficient path, $p = 200$.




Lasso training, test, and CV errors, $p = 200$.

| Selection | $\lambda$ | $f(\lambda)$ | Nonzero | Kept of $x_1$–$x_4$ | Test MSE |
|---|---|---|---|---|---|
| Test-grid oracle | 0.3051 | 0.00582 | 9 | 1,2,3,4 | 4.0100 |
| 5-fold CV minimum | 0.2615 | 0.00644 | 10 | 1,2,3,4 | 4.0238 |
| 5-fold CV 1-SE | 0.9328 | 0.00367 | 3 | 1,2,3 | 5.2401 |
| 10-fold CV minimum | 0.2242 | 0.00704 | 13 | 1,2,3,4 | 4.0493 |
| 10-fold CV 1-SE | 0.7996 | 0.00403 | 3 | 1,2,3 | 4.9054 |

**Rank and reference convention.** At $p = 200$, centering $n = 200$ observations gives

$$\operatorname{rank}(\widetilde{X}) \le 199.$$

The unpenalized least-squares slope solution is therefore nonunique. The script uses the minimum-Euclidean-norm solution

$$\widehat{b}_{\mathrm{LS}} = \widetilde{X}^{+}\widetilde{Y}$$

as the explicit reference for the Lasso shrinkage fraction.

Different least-squares slope solutions can have different $L_1$ norms, so this convention affects the numerical fraction. It cannot, however, push the fraction above 1. Let $L(b) = \frac{1}{2n}\|\widetilde{Y} - \widetilde{X}b\|_2^2$ and let $b_{\mathrm{LS}}$ be *any* least-squares solution. For $\lambda > 0$, optimality of $\widehat{b}_\lambda$ and of $b_{\mathrm{LS}}$ give

$$L(\widehat{b}_\lambda) + \lambda\|\widehat{b}_\lambda\|_1 \le L(b_{\mathrm{LS}}) + \lambda\|b_{\mathrm{LS}}\|_1, \qquad L(b_{\mathrm{LS}}) \le L(\widehat{b}_\lambda),$$

so $\|\widehat{b}_\lambda\|_1 \le \|b_{\mathrm{LS}}\|_1$ and $f(\lambda) \in [0, 1]$ whenever $\|b_{\mathrm{LS}}\|_1 > 0$.

The fraction is also monotone. For $0 < \lambda_1 < \lambda_2$, write $P_i = \|\widehat{b}_{\lambda_i}\|_1$. Optimality at each penalty gives

$$L(\widehat{b}_{\lambda_1}) + \lambda_1 P_1 \le L(\widehat{b}_{\lambda_2}) + \lambda_1 P_2, \qquad L(\widehat{b}_{\lambda_2}) + \lambda_2 P_2 \le L(\widehat{b}_{\lambda_1}) + \lambda_2 P_1.$$

Adding yields $(\lambda_2 - \lambda_1)(P_2 - P_1) \le 0$, so $P_2 \le P_1$: $f$ is nonincreasing in $\lambda$. Individual coefficients need not be monotone, but the total $L_1$ norm is. The script checks both properties; no violation occurred.

**Small-penalty limit.** At $p = 200$ the fraction reaches only about 0.31 at the end of the grid, unlike $p = 20$ and $p = 100$, where it approaches 1. To see why, let $\mathcal{B}$ be the set of least-squares solutions. The bound above gives $\|\widehat{b}_\lambda\|_1 \le \min_{b \in \mathcal{B}} \|b\|_1$, so the path is bounded and has accumulation points as $\lambda \downarrow 0$. Moreover $L(\widehat{b}_\lambda) \le \min L + \lambda\|b_{\mathrm{LS}}\|_1 \to \min L$. Hence every accumulation point $\bar{b}$

- minimizes the squared loss, i.e. $\bar{b} \in \mathcal{B}$, and
- has minimum $L_1$ norm within $\mathcal{B}$.

If the minimum-$L_1$ least-squares solution is unique (as it is for a design in general position), then $\widehat{b}_\lambda$ converges to it; without uniqueness, only the accumulation-point statement holds. The minimum-$L_2$ reference $\widetilde{X}^{+}\widetilde{Y}$ need not minimize the $L_1$ norm over $\mathcal{B}$, so the limiting fraction can be strictly less than 1. The value 0.31 is computed at the positive endpoint penalty $10^{-4}\lambda_{\max}$, so it is not itself an estimate of the exact limiting fraction.

At $p = n = 200$, very weak ridge penalties approach interpolation of the training data, so the training error approaches zero while the test error becomes very large. The ridge coefficient-path figure therefore adds a detail panel that excludes the near-interpolation penalties. Positive ridge penalties remain well defined even within CV folds whose training sample size is smaller than the number of covariates.

Changing $p$ changes the realized observations, even when the seed is the same. Thus comparisons across dimensions are not paired comparisons on identical responses, although the generating mean formula remains unchanged.

**Paired comparisons.** Ridge minus Lasso squared test errors, as in Exercise 1.17:

$p = 100$:

| Selection | Ridge | Lasso | Ridge − Lasso | SE | z |
|---|---|---|---|---|---|
| Test-grid oracle | 4.538 | 4.437 | 0.101 | 0.167 | 0.606 |
| 5-fold CV minimum | 4.554 | 4.675 | -0.121 | 0.200 | -0.606 |
| 5-fold CV 1-SE | 6.327 | 6.227 | 0.099 | 0.250 | 0.398 |
| 10-fold CV minimum | 4.554 | 4.637 | -0.084 | 0.197 | -0.424 |
| 10-fold CV 1-SE | 5.574 | 5.835 | -0.261 | 0.253 | -1.030 |

$p = 200$:

| Selection | Ridge | Lasso | Ridge − Lasso | SE | z |
|---|---|---|---|---|---|
| Test-grid oracle | 5.481 | 4.010 | 1.471 | 0.435 | 3.384 |
| 5-fold CV minimum | 5.913 | 4.024 | 1.889 | 0.502 | 3.760 |
| 5-fold CV 1-SE | 5.751 | 5.240 | 0.511 | 0.347 | 1.472 |
| 10-fold CV minimum | 5.816 | 4.049 | 1.767 | 0.473 | 3.734 |
| 10-fold CV 1-SE | 5.751 | 4.905 | 0.846 | 0.350 | 2.415 |

**Findings for $p = 100$.** The methods had similar test-grid minima (ridge 4.54, Lasso 4.44), and minimum-CV tuning slightly favored ridge on the observed test sample (4.55 versus 4.64–4.68), with every paired difference within about one standard error. On this realization there is therefore no clear evidence that either method predicts better; both CV-tuned fits sit roughly 0.5 above the linear floor of 4.06.

**Findings for $p = 200$.** Minimum-CV Lasso achieved test MSEs of 4.02–4.05 while keeping 10–13 nonzero slopes including all of $X_1, \ldots, X_4$; ridge achieved 5.82–5.91. The Lasso values lie slightly below the linear floor of 4.06, which is possible because test MSE is a noisy estimate of risk; they indicate small excess risk, not performance better than the best linear predictor. The paired differences favor Lasso relative to their estimated conditional test-sample uncertainty ($z \approx 3.7$), without establishing superiority across repeated training samples. This is consistent with the sparse population linear target $(1, c, c, c, 0, \ldots, 0)$: Lasso can set irrelevant coefficients exactly to

zero, whereas ridge shrinks without selecting variables. The one-SE rule slightly improved ridge's observed error (5.75) but worsened Lasso's (4.91–5.24), whose one-SE fits keep only 3 slopes and drop $X_4$. Weakly penalized ridge approached interpolation and showed pronounced test-error inflation, and CV chose a smaller ridge penalty than the test-grid benchmark (0.39–0.44 versus 0.99).